GEOLOGICAL HISTORY OF THE ISRAEL COASTAL PLAIN LONGITUDINAL FAULT ZONE; THE BASEMENT CONFIGURATION INDICATED BY MAGNETO-TELLURIC MEASUREMENTS

by

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ABSTRACT

An ancient hinge-line, running in a N-S direction occurs in the subsurface of the Coastal Plain. A longitudinal fault zone and stratigraphic changes are indicated along this hinge-line. A magneto-telluric profile was surveyed across the Coastal Plain fault zone in the Heletz-Ashqelon area. It is inferred that the Precambrian basement is displaced from a depth of 6.3 - 8.5 kms. in the eastern upthrown block, to a depth of more than 10.6 - 11.6 kms. in the western downthrown block. It is estimated that in the eastern block, Carboniferous sediments unconformably overlie the basement in stratigraphic similarity to the Negev and Sinai, where the Cambrian (?), Ordovician, Silurian and Devonian are missing. Subdivision of the sedimentary column in the eastern block is estimated as follows: Carboniferous-Permian (the Negev Group equivalent) 1.5 - 2 kms.; Permo-Triassic (the Ramon Group equivalent) 1.5 - 2 kms.; Jurassic, 2 - 3 kms.; Cretaceous, approximately 1 km.; Tertiary and Quaternary, 0.6 km. The western sedimentary column may be divided as follows: Infra-Cretaceous sediments - more than 8 kms., Cretaceous approximately 1.8 kms., Tertiary and Quaternary 1.8 kms.

There are differences between the two sides of the fault in lithology, environments of deposition, and thickness of formations of the Jurassic, Cretaceous and Neogene. The fault is an ancient hinge-line which was active at least in Late Jurassic - Cretaceous times and during the Neogene. During the Senonian to Oligocene time interval, when folding movements occurred, the fault was not active. Tectonic activity of the fault is associated with volcanism and erosion in the upthrown eastern block. There does not appear to be any lateral shift along the fault.

INTRODUCTION

Aim

Subsurface information on the Coastal Plain of Israel, based on oil well data and geophysical surveys, throws new light on the stratigraphy and structural geology of Israel. From this information, it was established that both in the western Coastal Plain and in the Mediterranean shelf, a sequence of shale and marlstone occurs, which is 3 - 4 kms. in thickness and which ranges in age from Cretaceous to Quaternary. In the eastern Coastal Plain, and the inland area of Israel, however, there is a variable composition of the time equivalent sedimentary column, which includes carbonates, chalks, marlstones, sandstones etc. The boundary between the eastern and the western sedimentary provinces seems to coincide with a longitudinal fault zone in the Coastal Plain.

The Coastal Plain of Israel, between Mount Carmel and the Gaza area, is a narrow belt of plains and hills, parallel to the Mediterranean coast. The belt is 5 - 30 kms. wide and 0 - 150 m. high (Fig. 1). In the subsurface, the Coastal Plain is divided into two structural blocks. South of Tel-Aviv, in the southern Coastal Plain, the boundary between the two blocks, is a line which extends in a N-S direction, parallel to the Mediterranean coast, and approximately in the middle of the Coastal Plain belt. In the northern Coastal Plain, this boundary is in the shelf area, very close to the sea shore. This boundary is a longitudinal fault zone, the western blocks of which are downthrown.

The purpose of the present study is to review the stratigraphy and structural geology of the Coastal Plain, and to point out that since the Jurassic the cause for structural and sedimentary separation is an ancient hinge-line in the Coastal Plain. This hinge-line is considered to be a fracture of the basement.

In order to prove this hypothesis, a preliminary survey of a magneto-telluric profile, perpendicular to the longitudinal fault lines, was carried out (Fig. 2). The Heletz-Ashqelon area was chosen because of abundant available well data. The magneto-telluric method was selected due to deep penetration of measurements it affords. These measurements might indicate the basement configuration of the Coastal Plain.

The longitudinal fault zone of the Coastal Plain - previous works

Picard (1959) named the western downthrown block in the Ashqelon area the "Ashqelon Trough". Gvirtzman (1969a) mapped the fault zone in a series of 1:100.000 subcrop geological maps. According to these maps, the fault zone is composed of several steps downthrown to the west, and is associated with minor transversal faults, running in a WNW direction (Fig. 3). The vertical displacement of the longitudinal faults is several hundreds of meters, and the cumulative displacement of the step faults, according to the base Saqiye datum, is approximately 2 kms. Vertical displacement of the transversal faults is several tens of meters.

Mapping of the faults is more definite in the Helez-Ashqelon area because of good well control. In other parts of the Coastal Plain, location of the faults is tentative, and based on interpolation between wells spaced several kilometers apart. Using gravity and seismic data, Ginzburg (1960) mapped a N-S major fault west of the Petah-Tiqva structure. This fault is in fact part of the fault zone discussed in the present study. Also using seismic data, Ginzburg and Amitai (1967), found two parallel longitudinal faults forming steps in the Ashqelon area. Klang (1967), mapped several seismic reflecting horizons of the Gan Yavne area, and showed two parallel steep gradients in a N-S direction. Gvirtzman (1969a) interpreted these two lines as part of the longitudinal fault zone of the Coastal Plain. Greenfield and Bein (in preparation) reinterpreted the seismic data of the Gan Yavne area, and found new evidence for the existence of the above mentioned faults. Fig. 3 shows the Miocene faults.

The gravity map of the Coastal Plain (Ginsburg, 1960, and new additional data by the Institute for Petroleum Research and Geophysics) shows a steep gradient of decreasing gravity towards the west in the Coastal Plain. This gradient does not specifically indicate the location of the longitudinal fault or fault system, but principally the shape of the fill of the Saqiye Group. The main density contrast occurs between the Saqiye marlstones, and the underlying limestone (Ginsburg, 1960).

The aero-magnetic map of the Coastal Plain (Domzalski, 1967) also indicates a steep gradient of the total magnetic field towards the west (Fig. 7). In his interpretation, however, Domzalski did not suggest a longitudinal fault in the Coastal Plain.

Stratigraphy of the Heletz-Ashqelon area

In order to understand the geological background as well as the present interpretation of magneto-telluric measurements, the stratigraphic column is reviewed according to well data. The review mostly refers to the Heletz-Ashqelon area, as information from that area is particularly abundant. Nevertheless the following description of the main lithostratigraphic units, from top to bottom, includes the entire Coastal Plain.

1. "Kurkar" Group - (Pleistocene). Composed of calcareous sandstones exposed over the entire Coastal Plain. Thickness ranges from 120m in the east to 160m in the west (Gvirtzman, 1969a, 1969b).

2. Saqiye Group - (Late Eocene to Early Pleistocene). Composed of marine marlstone, and distributed throughout the Coastal Plain. To the east, near magneto-telluric stations 38 and 39 (Fig. 2), thickness is 100-300m. There is a distinct thickening towards the west, and maximum thickness of 1800m occurs at the sea coast (Gvirtzman, 1969a, 1970).

3. Hashephela Group - (Senonian to Middle Eocene). Sediments of the Group were never deposited in the center of the Heletz-Ashqelon area. A thin veneer of Middle Eocene chalks covered the center of the area, but was eroded west of it, prior to the deposition of the Saqiye Group. Only minor remains of Hashephela Group still occur in the margins of the Heletz-Ashqelon area (Fig. 5) (Gvirtzman, 1969a, 1970).

4. Judea Group - (Albian-Turonian). The Judea Group was only deposited in the eastern part of the Coastal Plain, in the area of magneto-telluric stations 38 and 39 (Fig. 5). The sediments are composed of dolomites and limestones which were deposited in a shallow shelf, and are associated with rudist reefs. Thickness is approximately 900m. Towards the west, a facies change into marlstones and chalks of the Talmey-Yaffe Formation occurs. An interfingering zone between the Judea and Talmey Yaffe sediments was noticed (Fig. 5), (Gvirtzman and Buchbinder, 1966; Arkin and Hamaoui, 1967; Gvirtzman, 1970; Bein, 1970).

5. Talmey Yaffe Formation - (Albian-Cenomanian, ? Turonian). This formation was only deposited in the western Coastal Plain, and is composed of marlstone and chalks which were deposited under deeper marine conditions simultaneously to the Judea Group. Thickness is approximately 900m (Gvirtzman and Buchbinder, 1966; Cohen, 1969; Gvirtzman, 1970; Bein 1970, 1971).

6. Heletz Formation - (Neocomian). The Formation is composed of sandstones, shale, dolomites and limestones, and is only distributed in the eastern Coastal Plain. Sands of the formation form the reservoir rock of the Heletz-Kokhav oil fields. The Heletz Formation was deposited in near-shore environments having both continental and marine elements. The formation changes lateraly and downwards into the Gevar'Am Formation which was deposited under deeper marine conditions; its thickness is approximately 250m (Cohen, 1969; Shenhav, 1970).

7. Gevar'Am Formation - (Neocomian). The Formation is composed of dark shale rich in organic matter and hydrocarbons. The Formation is considered the source rock of the Heletz-Kokhav oil fields. It was deposited under deeper marine conditions; thickness ranges from 100m in the east to nearly 1,000m in the west (Cohen, 1969; Shenhav, 1970; Shenhav and Shoresh, in preparation).

8. Jurassic rocks - Composed mostly of limestones, dolomites and some shale beds, and mostly deposited under shallow marine conditions. Maximum known thickness is 2,500m, penetrated in the Heletz 22 well (Picard and Eliezri, 1964; Derin, 1966; Cohen, 1969; Barzel and Friedman, 1970). According to B. Derin (personal note) the bottom of the Heletz 22 well is close to the Jurassic-Triassic boundary. No deeper formations were penetrated in the Coastal Plain.

MAGNETO-TELLURIC MEASUREMENTS

The magneto-telluric method

The magneto-telluric method, based on Cagniard's theory (1953) of micro-pulsations of plane electromagnetic waves incident on the earth, involves simultaneous surface measurement of the horizontal components of the magnetic and electric fields.

Cagniard showed that quantitative information concerning the electrical properties of the subsurface can be obtained for horizontally layered earth, where each layer is homogeneous and isotropic. He equally demonstrated that resistivity can be found as a function of depth if the apparent resistivity ρ_a is known as a function of period T.

Apparent resistivity is computed from:

$$\rho_a = 0.2\,T \left| \frac{E_y}{H_x} \right|^2$$

where

ρ_a = apparent resistivity in ohm meters

T = period of the magneto-telluric pulsation in seconds

E_y = horizontal component of electric field in direction y, in mV/Km.

H_x = horizontal component of magnetic field in a perpendicular direction, x, in gamma (γ).

In the local application of the magneto-telluric method one horizontal component of the electric field (Ey) and one of the magnetic field (Hx) were measured (Fig.

The magnetic system consists of a large induction coil whose signal is amplified by a three-stage amplifier with a calibrated gain. Included in the amplifier are notch filters for 50 Hz and 150Hz and a system of band pass filters (combination of high-cut and low-cut filters) for the pre-filtering of signals. The telluric system (measuring the electric field) consists of two porous-pot electrodes planted in the ground, about 600m apart connected by insulated leads to the same amplifier as that used for the magnetic system. This recording system was built by "Centre d'Etudes Geophysiques", France.

Data Analysis

Eight stations were chosen for field measurements, the locations of which are shown in Fig. 2. The stations are located in a line perpendicular to the faults of the Coastal Plain. Data obtained at station 36 could not be used, and this station has not been considered in the discussion below. Data obtained at station 32 shows very low resistivity values at depths of about 2.5 km probably due to the penetration of sea water. These values do not represent the lithology of the strata, and this site too, was not included in the following discussion.

Records obtained directly, or by play-back of the magnetic tapes for the selected periods, were analysed individually, and all the well correlated magnetic and telluric events were considered. From the amplitude ratio of these events the value of the apparent resistivity ρa was calculated for a varying period T.

Values of ρ_a versus $\sqrt{T}$ were plotted on log-log coordinates, and using master curves, an earth model of resistivities and thickness was calculated. Using a multilayer case computer program, this proposed model can be verified by reconstructing the corresponding theoretical curve.

Results

Table 1 shows results of magneto-telluric measurements of 6 stations along a profile in the Heletz-Ashqelon area, perpendicular to the fault zone (see Fig. 2). The results show depth soundings of intervals measured in meters and characterised by resistivity values in ohm-cms. Fig. 5 shows a geological cross section (horizontal= vertical scale) along the measured profile, indicating main lithostratigraphical units down to the Jurassic.

At the bottom of Fig. 5 the results of the magneto-telluric measurements are shown along the same profile. The resistivity values are plotted on a log scale. The magneto-telluric units are designated by letters as magneto-telluric units. Table 2 shows the properties of the magneto-telluric units and their possible lithostratigraphic equivalents. From the Cross-sections of Fig. 5, there appears to be good correlation between the following lithostratigraphic units: "Kurkar" Group, Saqiye + Talmey Yaffe and Gevar'Am formations and the magneto-telluric units, A, C and D. Table 3 shows details of these correlations. Although there are differences between the depth and boundary of the lithological and geophysical units, a statistical analysis of correlation between the 4 contacts obtained shows a correlation coefficient as high as 0.96 (Fig. 6).

DISCUSSION AND SYNTHESIS

Significance of the magneto-telluric units

For the purpose of the present discussion, a magneto-telluric unit is defined by its resistivity and thickness ranges and by its position in the sequence. Table 2 is a summary of the properties of the defined magneto-telluric units. Fig. 5 shows the same units in cross section. The following are complementary remarks to Table 2 and Fig. 5.

Unit A - The Unit is exposed, its upper part being within the weathered zone or the "conductive overburden" (Fig. 4). In spite of relatively low resistivity values, the upper part of the unit makes up the fresh water aquifer of the Coastal Plain. This result needs further study.

Unit B - The unit is only found in the eastern part of the section (stations 38 and 39, Fig. 5). The unit differs from unit C, which forms the lateral extention of B to the west, Resistivity values of unit B are double than those of Unit C (Table 2). Unit B has a thickness of 4-7 kms., while Unit C is only 2 kms. in thickness. The underlaying units also differ; B overlies F, and C overlies D. D and F are different units with different properties. The wells of the Heletz-Kokhav field penetrate the interval of the upper part of Unit B. This depth interval includes the approximate 900m of the Judea Group Carbonates, the approximate 250m of the Heletz Formation sandstones, the approximate 100m of the wedged-out Gevar'am Formation shale and the approximate 2.5 kms. of Jurassic carbonates which were penetrated in the Heletz 22 well. The total section mentioned above (approximately 3.75 kms), is mostly composed of carbonates. The approximately 350m of sandstone and shale does not change the geophysical properties of the entire Unit B sequence. The lower part of Unit B is older than Jurassic, but is probably of similar lithology.

Unit C - This unit is the lateral westwards extention of the upper part of Unit B. It corresponds to the marlstones of the Saqiye Group, and the Talmey Yaffe Formation.

Unit D - The unit is equivalent to the Gevar'Am Formation. Extremely low resistivity values here are probably due to the overpressured conditions of the Gevar'Am shale (see Fertl and Timko, 1970). There appears to be a high pressure anomaly in the Gevar'Am shale, in which the formation pressure exceeds the expected hydrostatic pressure.

Unit E - This unit is the thickest in the studied area. The bottom of the unit was not reached at 8 kms. The upper part of the unit corresponds to the Jurassic carbonates which were penetrated in wells in the Ashqelon area. The entire unit is apparently of carbonates.

Unit F - The unit is only known in the eastern part of the profile. It is possible, however, that the unit may underlie unit E in the west. Resistivity values of F are low.

Unit G - Due to its high resistivity, this unit is presumed to be the Precambrian basement. In other magneto-telluric measurements in Israel, resistivities of Precambrian basement rocks were found in the order of magntiude of 18-46 x 10^3 ohm-cm. Measurements in the Timna' area in the Negev (Klang, 1970) showed the resistivity of the Precambrian crystalline basement to be 18 x 10^3 ohm-cm.

In the Znifim-Paran-Meishar area in the Negev (Klang and Weissbrod, in preparation) values of 19-46 x 10^3 ohm-cm were recorded at a depth that corresponds to the so-called "Infra-Cambrian Znifim Arkose" (Weissbrod, personal note, see also Weissbrod, 1969a). The Znifim Arkose, a thick sequence in the northern Negev, composed of degradated granitic particles, is part of the so-called "Precambrian Basement" of the Arabo-Nubian massif, which is mainly composed of igneous and metamorphic rocks. In Israel, therefore,

magneto-telluric data points to the Znifim Formation as part of the Precambrian basement complex. When thus classified, the prefix "Infra-Cambrian" which was previously given to the formation, can be disregarded. In the other magneto-telluric measurements that were carried out in Israel, high resistivities of this nature were not found within the overlaying sedimentary column (except in the salt-beds of the Dead Sea area). Consequently, the G unit of the Heletz area is correlated with the Precambrian basement. This type of unit was not penetrated in stations at the western part of the profile.

Location of a hinge-line and configuration of the basement

For the purpose of the present discussion, an ancient hinge-line is defined, as a line along which fault displacement, sharp changes in thickness of sediments and/or sharp changes in environments of deposition have occurred. A hinge-line of this nature is probably the expression of an active fracture in the earth's crust, known to cause rejuvenation of ancient faults. The latter process brings about paleogeographic features such as shelf edges, fault scarps, etc. These paleogeographic features form boundaries between sedimentary provinces and different environments of deposition. Thus, the term "hinge- line", in this discussion, indicates a tectonic and/or a depositional hinge.

Magneto-telluric measurements in the Heletz-Ashqelon area (Fig. 5) indicate a major lateral change in the nature of the magneto-telluric units between stations 37 and 38. The geological cross section also shows a lateral change in the upper sedimentary column, mostly between stations 37 and 38. The Heletz Formation passes laterally to the Gevar'Am Formation, the Judea Group passes to the Talmey-Yaffe Formation and from a negligible

thickness of the Saqiye Group, there is a change into a thick section of the Group.

In stations 38 and 39, the depth of the Precambrian basement is estimated to be in the range of 6.3 - 8.5 kms (Tables 1 and 2). There is a gradient to the west between these two stations, which might express a fault or a regional dip. The Precambrian basement was not penetrated in the next four stations but it is deeper than 10.6 kms in station 37 and deeper than 11.6 kms in station 33 the westernmost station. The fact that the Precambrian basement was not penetrated in station 37, might further confirm the idea that the main fault zone is located between station 37 and 38. This assumption is equally supported by an aero-magnetic profile. The aero-magnetic map here under discussion (Domzalski, 1967) indicates two steps in the profile of the observed total magnetic field along the direction of the magneto-telluric survey (Fig. 7). The main fault zone, between stations 37 and 38 is a boundary between the steps in the magnetic profile.

In conclusion, the above mentioned changes are expressed by the following features: a lateral change in the magneto-telluric properties of the sedimentary column; a lateral stratigraphic change in the upper sedimentary column; a fault zone, and a cumulative displacement of the Precambrian basement of more than 3-5 kms; and a steep gradient in the total magnetic field along the profile. Most of these features are focussed between stations 37 and 38.

It thus appears that the fault zone is actually an old hinge-line which follows a fracture in the earth's crust. This fracturing of the crust was active several times during the geological history of the Coastal Plain. This hypothesis is discussed in the following paragraphs. The estimated location of the Coastal Plain fault zone or hinge-line is shown in Fig. 8.

Possible age of the deeper parts of the sedimentary column of the Coastal Plain

In wells of the northern Negev, as well as in outcrops of southwest Sinai, Carboniferous rocks unconformably overlay "Infra-Cambrian" arkose rocks and/or thin marine Cambrian sediments. Carboniferous rocks occasionally also directly overlay the crystalline basement. In the Negev as well as in Sinai there is a huge sedimentary gap, in which Ordovician, Silurian and Devonian sediments are lacking. These lower Paleozoic sediments are only found in north-west Saudi Arabia and in south Jordan. In the Negev and Sinai the carboniferous sediments overlay widespared truncated relief (Weissbrod, 1969a, 1969b, 1970).

For the most part, the thickness of the sedimentary cover which overlies the truncated old relief in the northern Negev may be divided as follows: 800m of the Carboniferous - Permian Negev Group (Weissbrod, 1969a), 800m of the Permo-Triassic Ramon Group (Druckman, 1966) and 1250m of Jurassic Arad Group (1000m from bottom of the section to the top of Zohar Formation according to Goldberg, 1970, and approximately 250m of the Jurassic Kidod, Be'er Sheva' and Haluza formations). The above mentioned groups thicken towards the north-west, in the general trend from the Arabo-Nubian massif into the old Thetis ocean. The rate of thickening towards the Heletz area in the Coastal Plain may be estimated according to the thickening of the Jurassic sequence. In the Ziqlag 2 well, between the northern Negev and the Coastal Plain, 1620m of Jurassic were penetrated (Fisher and Druckman, 1968). In the Heletz 22 well, 2500m of Jurassic were penetrated. According to B. Derin (personal note), the bottom of the sequence is very close to the Jurassic-Triassic boundary. Thus it may be concluded that the Jurassic of the Coastal Plain is twice thicker than the Jurassic of the Negev. A two fold thickening is

is also assumed for the Negev and Ramon group and the expected section in the Heletz area is 1600m of Carboniferous - Permian, 1600m of Permo-Triassic in addition to the 2500m of Jurassic. A total of 5.7 kms of Carboniferous-Jurassic sediments is estimated.

The magneto-telluric measurements in the eastern part of the profile (station 38 and 39) indicate similar thickness. In station 39, which is located near the Heletz 34 well and in which the top of the Jurassic is at 1760m depth, the total thickness of infra-Cretaceous sediments is 4,5 kms (from 1760m to 6300m, the top of the inferred Precambrian basement). In station 38, where the top of the Jurassic, by extrapolation and seismic data, is at 1800m depth, the total thickness of infra-Cretaceous sediments is 6.7 kms (from 1800m to 8500m, the top of the inferred Precambrian basement). The thickness range of the infra-Cretaceous sediments in stations 38 and 39 is 4.5-6.7 kms, which is close to the above mentioned estimated 5.7 kms of Carboniferous-Jurassic sediments. From the similarity of these estimates a working hypothesis may be deduced that the total infra-Cretaceous sediment in the Heletz area is also of the late Paleozoic to Mesozoic age. Lower Paleozoic sediments, which are absent in the Negev and Sinai (either due to non deposition or truncation) are probably also absent in the Coastal Plain belt.

Thickness and facies changes along the hinge-line

Infra-Cretaceous - For the purpose of the present discussion, Infra-Cretaceous is defined as Jurassic or older. However, it is probable, as assumed previously, that the Precambrian basement, in the upthrown block of the Coastal Plain, is overlain by Carboniferous rocks. The Infra-Cretaceous sediments in the east, therefore, are probably Carboniferous to Jurassic.

As mentioned previously, the thickness range of the infra-Cretaceous sediments in the eastern block is 4.5-6.7 kms. In the western down-faulted block, in station 37, the thickness of the infra-Cretaceous sediments (Unit E), exceeds 8 kms and the thickness in station 34 exceeds 8.5 kms. The thicker section of the infra-Cretaceous in the western down-thrown block is either due to the presence of sediments older than Carboniferous, or to a thicker accumulation of sediments dating from the Carboniferous to the Jurassic time interval.

There are some indications for facies changes in Jurassic sediments. There is no sharp change in lithology in Jurassic carbonates between the two sides of the hinge-line in the Heletz-Ashqelon area. Nevertheless, Barzel and Friedman (1970) found different environments of deposition of late Jurassic sediments between the carbonates of the Heletz belt, which were deposited in very shallow marine conditions, in a high-energy zone above the sea wave base, and those west of the Heletz belt, which, they assumed, were deposited under low-energy open sea conditions, beneath wave base. In the northern Coastal Plain, along the northern extention of the fault zone, there is a distinct difference in Jurassic rock on either side of the hinge-line. Friedman, Barzel and Derin (1971) found different environments of deposition in the Upper Jurassic rocks. In the eastern upthrown block of the fault zone, which is represented by the Caesarea 1, Carmel 1, Haifa 1, Asher 1 and 2, and Haifa Bay 1 bore holes, they found Jurassic sediments of shelf and shelf edge environments of deposition. In the western down-faulted block, which is represented by the off-shore Delta 1 bore hole, they found Jurassic sediments of basin margin (seawards, off the basin slope) environment.

Facies changes in Late Jurassic sediments might be connected with a displacement in the fault zone and with an accumulation of thicker sediments in the down-thrown block.

It is also possible that the few kilometers of thicker section in the western block, were accumulated during the Jurassic rather than over the entire pre-Cretaceous to post Precambrian time interval.

Lower Cretaceous - A facies change also occurs in Lower Cretaceous sediments on either side of the hinge-line. The Heletz Formation in the eastern block includes sediments of continental dune sands, tidal channels and lagoons, off shore marine sands as well as reef dolomites, carbonaceous shale and volcanic ash (Shenhav, 1970). The Formation passed laterally into the Gevar'Am Formation, which is mostly distributed along the western side of the hinge-line, The Gevar'Am Formation was deposited in open sea conditions (Cohen, 1969). Shenhav and Shoresh (in preparation) pointed out some evidence for a rapid rate of deposition of the Gevar'Am Shale, which explains the accumulation of more than 950m of the shale in the downfaulted block, while the Heletz Formation, which was deposited near the shore only has a thickness of 250m.

In the western down-thrown block an excess of sediments of approximately 0.7 km was deposited during lower Cretaceous times.

Albian-Cenomanian - In rocks of Albian to Cenomanian age, there is also a facies change on either side of the hinge-line. The Judea Group, in the eastern block is composed of shallow marine carbonates, rudist reefs, and platform limestones and dolomites (Bein, 1971). To the West, there is a facies change through a zone of interfingering, into open sea marlstones (Gvirtzman and Buchbinder, 1966; Cohen, 1969; Gvirtzman, 1970; Bein, 1970, 1971).

There is no distinctive thickness change between the Judea and the Talmey-Yaffe units in the Heletz Ashqelon area. Nevertheless, 2.6 kms of Albian to Cenomanian

sediments were penetrated in the Item 1 off-shore well, which faces the northern Coastal Plain, and which represents the down-thrown block. However, thickness of the Judea Group, in the upthrown block of the northern Coastal Plain is only 700-900m.

Neogene - In sediments of the Saqiye Group (Late Eocene to Early Pleistocene) there is also a distinct facies change between the two sides of the hinge-line. In the western down-thrown block, marine conditions persisted almost throughout the entire time of deposition of the Saqiye Group (except during a relatively short break in the Miocene) and approximately 1800m of open marine globigerinal marlstones were deposited. The Ashqelon Member, and the Ziqim and Mavqi'im formations were only deposited in the western trough.

The eastern upthrown block was exposed to erosion during the early to middle Miocene. Along the hinge-line there was a shore belt in which the fault scarp formed a sea-shore cliff. The eastern block was an exposed inland plateau, where erosion channels were cut into the fault scarp (Gvirtzman, 1969a, 1970). The cumulative displacement of the fault zone during the Neogene was estimated by Gvirtzman (1970) as approximately 2 kms.

Rates of displacements - According to the above hypothesis it is possible to estimate rates of displacements in the Heletz-Ashqelon area. It is assumed that the "Infra-Cretaceous displacement" took place mostly in the late Jurassic times and is associated with the facies changes of the same time. The Lower Cretaceous and Albian-Cenomanian displacements seem to be a continuation of the same movement. The movement probably commenced during the late Jurassic and probably terminated during the late Cenomanian. The excess of thickness of Infra-Cretaceous sediments in the western block is more than 1.8-4.0 kms.

During the Lower Cretaceous, 0.7 kms of sediments were accumulated. No distinctive accumulation is known during the Albian - Cenomanian. As a possible result of Jurassic-Cretaceous movements, the total excess of sediments in the western block area, exceeds 2.5-4.7 kms. The Jurassic-Cretaceous displacement continued approximately over a period of 70 million years and therefore, rate of displacement is estimated at over 36-67m per one million years.

The Neogene displacement is estimated by Gvirtzman (1970) as 100m per one million years. A graphic summary of these estimations is presented in table 4.

Volcanism and erosion

Tectonic movements in the fault zone along the hinge-line area, both during the Late Jurassic-Cretaceous, and during the Neogene, appear to be also connected with volcanic activities and erosional phases in the Coastal Plain.

In Israel the Jurassic-Cretaceous movement was followed by volcanics. Cainozoic volcanic activities were described by Bentor (1957). Volcanic rocks from the Jurassic-Cretaceous boundary were found in bore holes of the northern Coastal Plain and northern Israel (Nachmias and Gvirtzman, 1967). The proximity of the Heletz Formation sediments to volcanic rocks is discussed by Shenhav (1970), while Cretaceous volcanism in the Carmel and Umm el Fahm areas is described by Sass (1957, 1968). Neogene volcanic activities in the Coastal Plain (National Park volcanics) which are associated with the fault zone now under consideration are described by Gvirtzman (1970).

Erosional phases, which are expressed by major unconformities, have been considered by several authors. The Jurassic-Cretaceous unconformity is discussed by

Eliezri (1964) and Cohen (1969). Greenfield and Bein (in preparation) suggested that the erosion channel in the Heletz-Ashqelon area, which cut the Jurassic rocks, and was filled by the Lower Cretaceous Gevar'Am Formation, was formed due to erosion in the rising upthrown block. The major Neogene uplifting, erosion and unconformity is discussed by Gvirtzman (1970) and illustrated in subcrop geological maps (Gvirtzman, 1969).

Normal rather than strike-slip fault

Differences in thicknesses and environments of deposition between two sides of a fault can either be explained, as suggested above, as an ancient hinge-line, or else as a strike slip fault. In a strike slip fault, the two sides of the fault do not represent identical points.

The differences in the Jurassic sediments on both sides of the Dead Sea rift have been previously explained to be a result of an ancient hinge-line (Bentor and Vroman, 1960) but were later considered to be the result of a strike-slip fault, in which both sides of the rift moved 105 km apart (for details see Freund et al. 1970).

There are indications that the Coastal Plain fault zone be an ancient hinge-line, since the Jurassic, or may be earlier, and not a left-lateral offset fault. The lateral facies change between the Gevar'Am and Heletz Formations is associated with an erosional channel cut into the Jurassic rocks and filled by lower Cretaceous marine sediments of the Gevar'Am formation, The channel of the Heletz area which was mapped in detail (Cohen, 1969; Shenhav and Shoresh, in preparation), extends across

Movements are associated with facies changes and thickness changes between the two sides of the hinge-line. They are also associated with volcanic activities and erosion which took place mostly in the eastern upthrown block; both the hinge-line, and the fault zone are shown in Fig. 8. Folding movements in the Coastal Plain, as well as over the entire area of Israel took place in one phase during the Senonian to Eocene, and in a rejuvenated phase during the Oligocene (Gvirtzman, 1970). Sediments deposited during folding, namely Hashephela Group in the first phase, and the Gat Member of the Beit Guvrin Formation as well as the Lakhish Formation in the second phase, do not differ in lithology, environments of deposition and thickness on either side of the hinge-line (Gvirtzman, 1970). It is highly probable that during the Senonian-Oligocene time interval, the Coastal Plain, on both sides of the hinge-line, was under compressional stress. The relief differences between the trough and the eastern uplifted blocks was smoothed by the end of the Toronian and folding took place.

A summary of the tectonic history of the Coastal Plain is shown in Table 4.

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TABLE 1

RESULTS OF MAGNETO-TELLURIC MEASUREMENTS AT 6 STATIONS

Station	Coord.	Elev., MSL	Interval, m.	Thickness, m.	Resistivity, Ω -cm.	Unit Symbol
33	104800/114270	22.0	0 - 290	290	750	A
			290 - 2,600	2,310	3,750	C
			2,600 - 3,640	1,040	27	D
			3,640 - >11,600	> 7,960	9,500	E
34	106800/114000	25.0	0 - 160	160	700	A
			160 - 2,000	1,840	2,100	C
			2,000 - 2,800	800	36	D
			2,800 - > 11,300	> 8,500	12,000	E
35	107500/111950	25.0	0 - 290	290	300	A
			290 - 1,450	1,160	3,000	C
			1,450 - 2,620	1,170	74	D
			2,620 - > 11,100	> 8,480	16,000	E
37	110054/110075	51.0	0 - 190	190	550	A
			190 - 2,100	1,910	4,500	C
			2,100 - 2,600	500	30	D
			2,600 - >10,600	> 8,000	10,000	E
38	111500/109400	75.0	0 - 320	320	550	A
			320 - 7,100	6,780	8,250	B
			7,100 - 8,500	1,400	200	E
			8,500		27,000	G
39	112750/109900	75.0	0 - 280	280	950	A
			280 - 4,500	4,220	6,650	B
			4,500 - 6,300	1,800	314	F
			6,300		? 25,000	G

TABLE 2

MAGNETO-TELLURIC UNITS (See also Figure 5)

Magneto-telluric unit	Resistivity range ohm - cm	Thickness range km	Inferred equivalent lithostratigraphic unit	Inferred lithology and remarks
A	300 - 950	0.160 - 0.320	"Kurkar" Group	Calcareous sandstones, exposed, the upper part is a fresh water aquifer
B	6,650 - 8,250	4.220 - 6.780	Judea Group (including a tongue of the Talmey Yaffe Formation), Heletz Formation (including wedge of the Gevar'Am Formation) Jurassic carbonates and older sediments.	Predominantly hard limestones and dolomites.
C	2,100 - 4,500	1.160 - 2.310	Saqiye Group and Talmey Yaffe Formation	Predominantly marlstones
D	27 - 74	0.500 - 1.170	Gevar'Am Formation	Shale, Very low resistivity-shale, probably over-pressured (see Fert and Timko. 1970).
E	9,500 - 16,000	7.960 - 8.500	Jurassic carbonates and older sediments	Probably hard carbonates
F	200 - 314	1.400 - 1.800	?	? Shale, Probably overpressured
G	25,000 - 27,000	?	Precambrian Basement	Probably igneous and metamorphic rocks

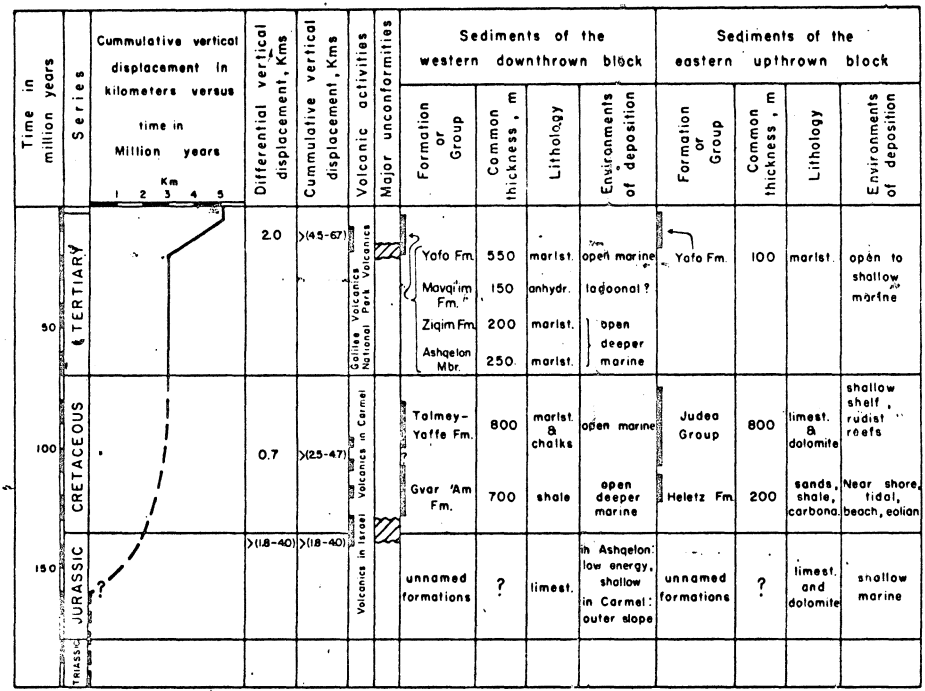
TABLE 3

CORRELATION BETWEEN DEPTH AND THICKNESSES OF LITHOSTRATIGRAPHIC UNITS AND MAGNETO-TELLURIC DATA

STATIONS / UNITS		33 (near Ziqim 1 well)		34		35		37 (near Shimon 1 well)		38		39 (near Heletz 34 well)	
Lithostratigraphic and Magneto-telluric		by geology*	by mag.-tell.	by geology*	by mag.-tell.	by geology*	by mag.-tell.	by geology*	by mag.-tell.	by geology*	by mag.-tell.	by geology*	by mag.-tell.
"Kurkar" Group, "A"	bottom, m.	167	290	140	160	130	290	91	190	125	320	125	280
	thickness, m.	167	290	140	160	130	290	91	190	125	320	125	280
Saqiye Group and Talmey-Yaffe Formation, "C"	bottom, m.	2775	2600	2450	2000	2300	1450	1614	2100	Only negligible thickness of Saqiye and No Talmey-Yaffe in this area.			
	thickness, m.	2608	2310	2310	1840	2170	1160	1523	1910	No Unit "C"			
Gevar'Am Formation, "D"	bottom, m.	3770	3640	3200	2800	2750	2620	1924	2600	Only negligible thickness of Gevar'Am Formation in this area.			
	thickness, m.	965	1040	750	800	450	1170	310	500	No Unit "D"			

* including by seismic data

Table 4



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Fig. 1

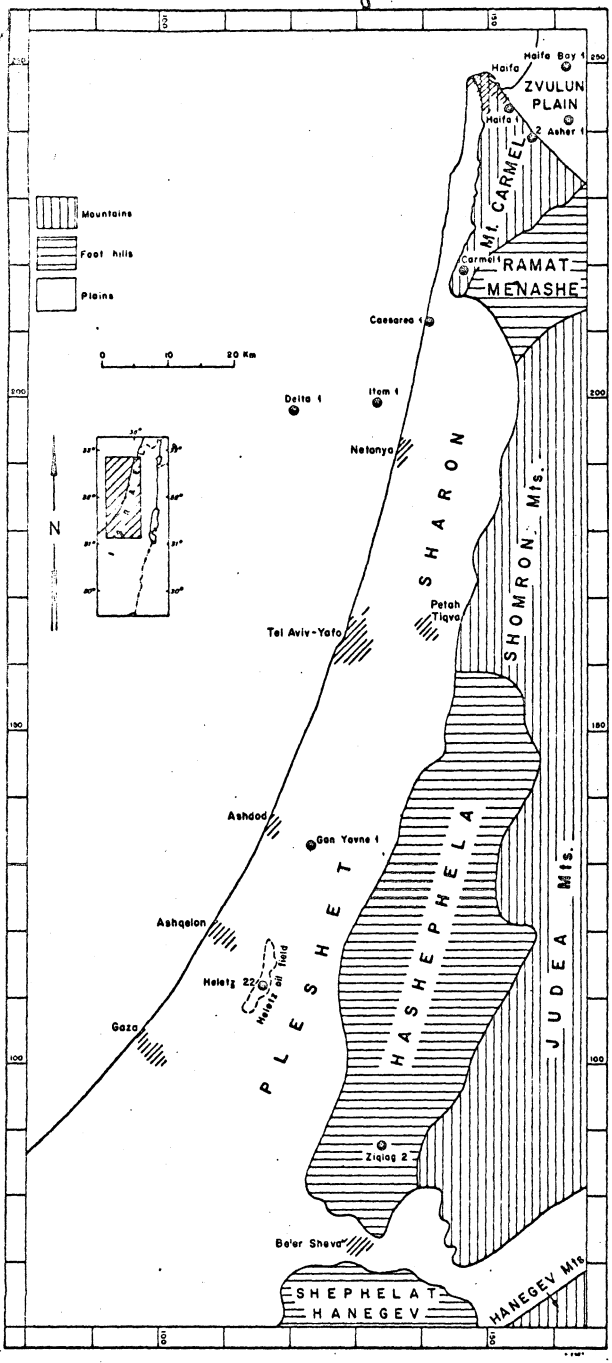

Haifa Bay 1
Haifa
ZVULUN PLAIN
Haifa 1
2 Asher 1
Mt. CARMEL
Carmel 1
RAMAT MENASHE
Caesarea 1
Mountains
Foot hills
Plains
0 10 20 Km
Delta 1
Itam 1
Netanya
SHARON
SHOMRON Mts.
Petah Tiqva
Tel Aviv-Yafo
N
Ashdod
Gan Yavne 1
Ashqelon
Heletz 22
Heletz oil field
Gaza
PLESHET
HASHEPHELA
JUDEA Mts.
Ziqlag 2
Be'er Sheva'
SHEPHELAT HANEGEV
HANEGEV Mts

Fig. 2

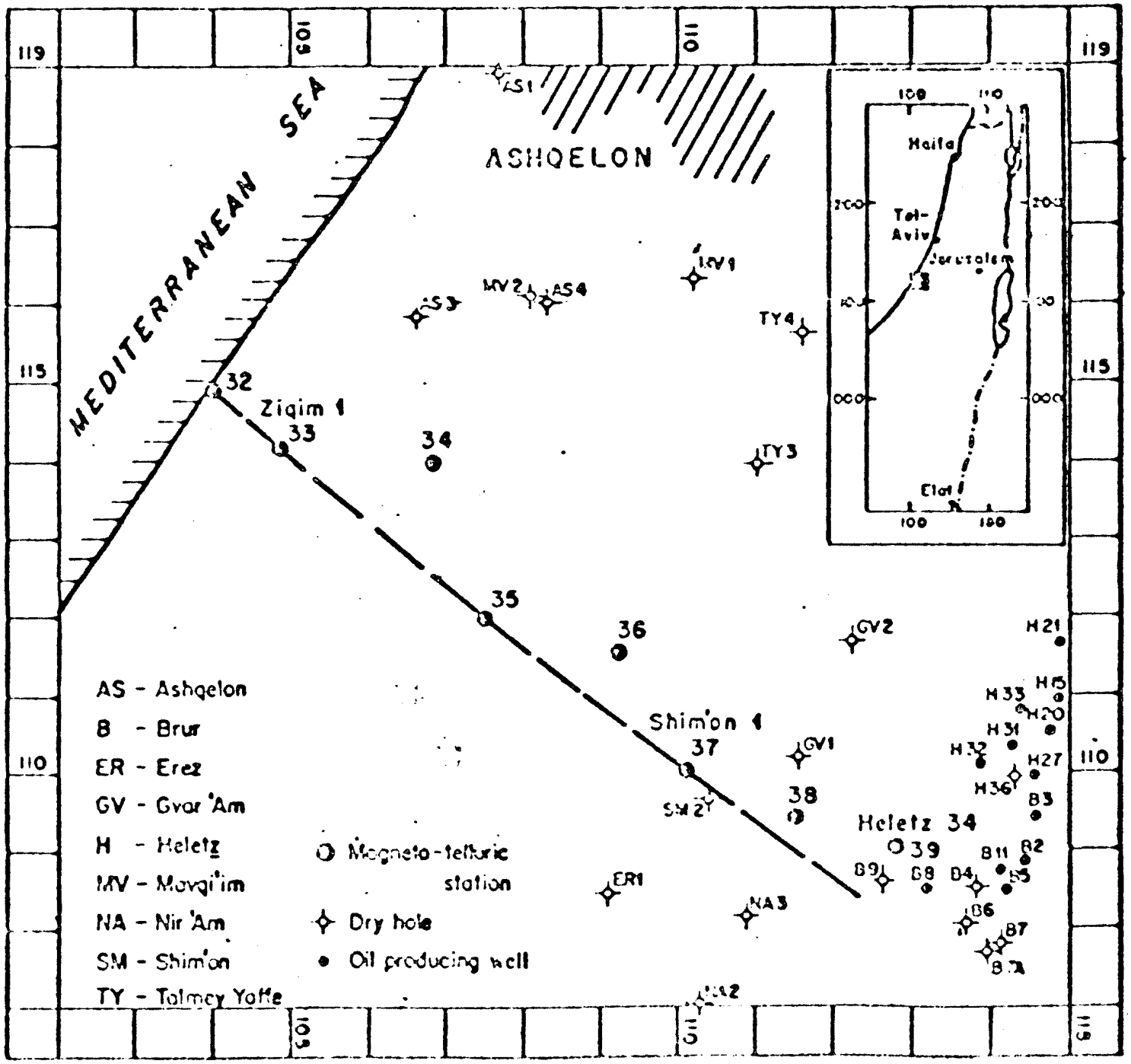
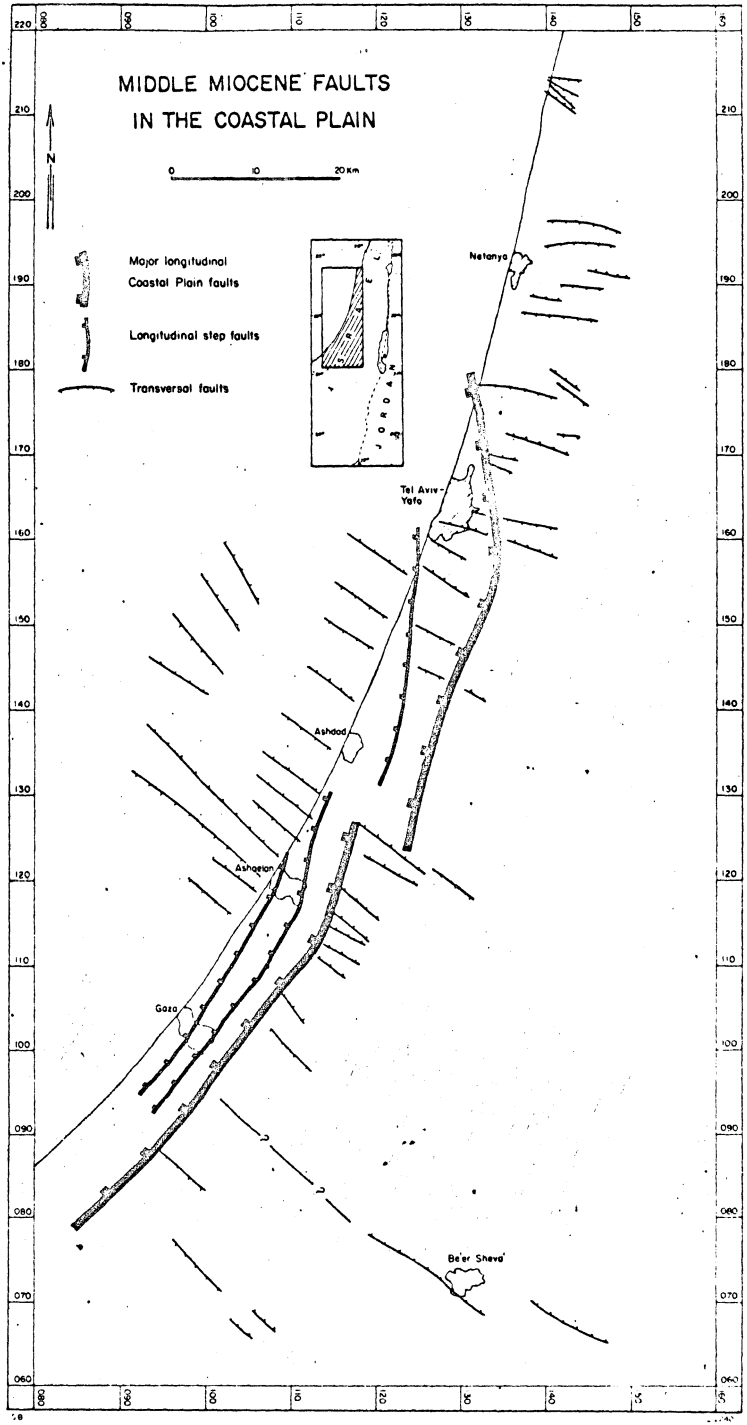

MEDITERRANEAN SEA
ASHQELON
AS1
MV2
AS4
MV1
TY4
TY3
AS3
32
Ziqim 1
33
34
35
36
GV2
H21
Shim'on 1
37
GV1
38
SM2
Heletz 34
39
ER1
NA3
AS2
AS - Ashqelon
B - Brur
ER - Erez
GV - Gvar 'Am
H - Heletz
MV - Mavqi'im
NA - Nir 'Am
SM - Shim'on
TY - Talmey Yafe
Magneto-telluric station
Dry hole
Oil producing well
Haifa
Tel-Aviv
Jerusalem
Eilat
100
110
200
000
119
115
110
105
111

Fig. 3


MIDDLE MIOCENE FAULTS
IN THE COASTAL PLAIN
N
0 10 20 Km
Major longitudinal Coastal Plain faults
Longitudinal step faults
Transversal faults
JORDAN
Netanya
Tel Aviv - Yafo
Ashdod
Ashqelon
Gaza
Be'er Sheva'

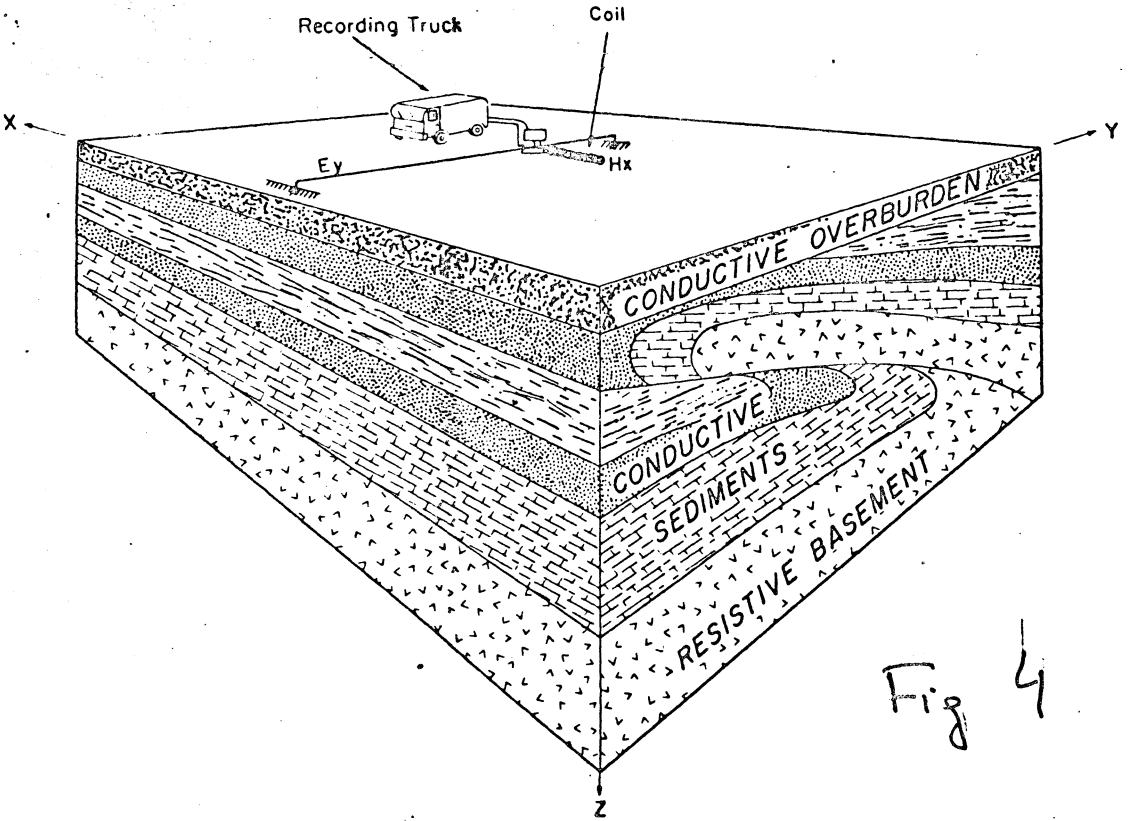
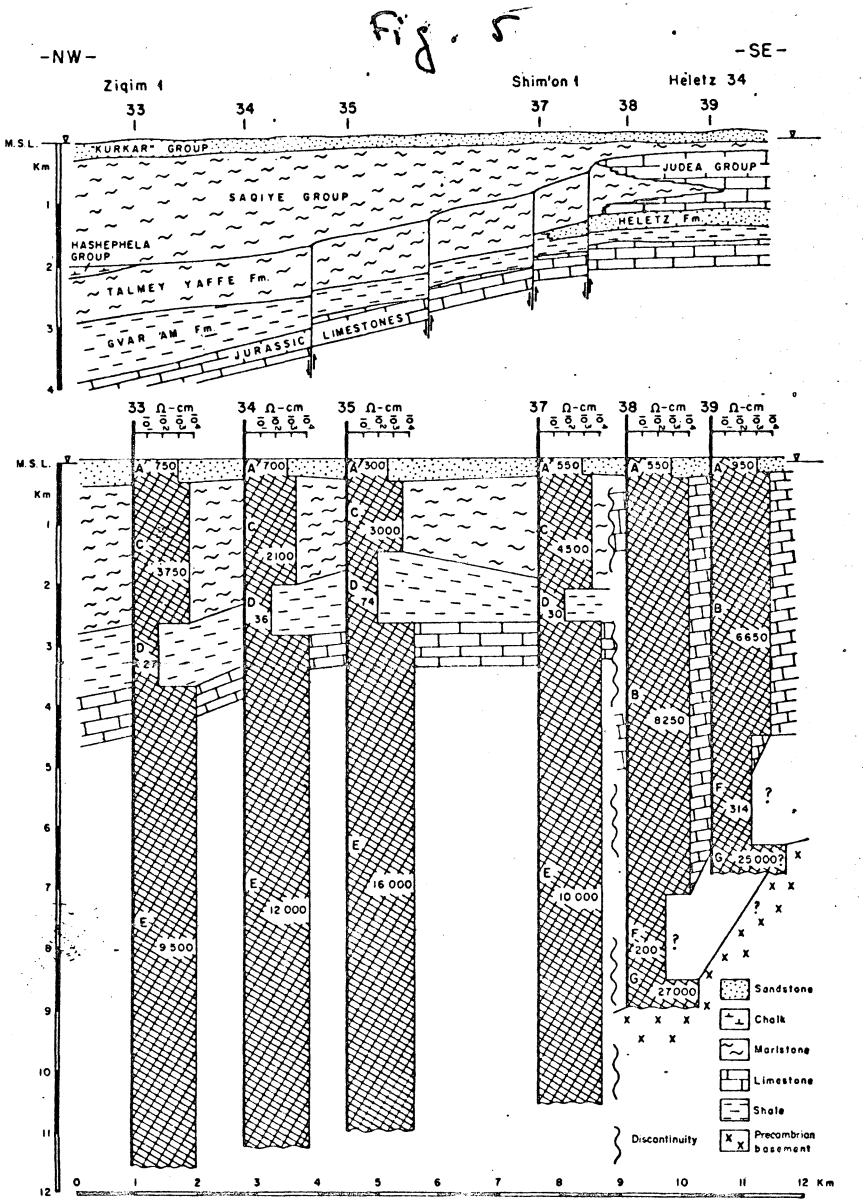

Recording Truck
Coil
X
Y
Ey
Hx
CONDUCTIVE OVERBURDEN
CONDUCTIVE
SEDIMENTS
RESISTIVE BASEMENT
Z

Fig 4

Fig. 5


-NW-
-SE-
Ziqim 1
Shim'on 1
Heletz 34
33
34
35
37
38
39
M.S.L.
Km
"KURKAR" GROUP
JUDEA GROUP
SAQIYE GROUP
HELETZ Fm.
HASHEPHELA GROUP
TALMEY YAFFE Fm.
GVAR 'AM Fm.
JURASSIC LIMESTONES
33 Ω-cm
34 Ω-cm
35 Ω-cm
37 Ω-cm
38 Ω-cm
39 Ω-cm
A 750
C 3750
D 27
E 9 500
A 700
C 2100
D 36
E 12 000
A 300
C 3000
D 74
E 16 000
A 550
C 4500
D 30
E 10 000
A 550
B 8250
F 200
G 27000
A 950
B 6650
F 314
G 25000?
Sandstone
Chalk
Marlstone
Limestone
Shale
Precambrian basement
Discontinuity
0 1 2 3 4 5 6 7 8 9 10 11 12 Km

D 2386

Fig. 6

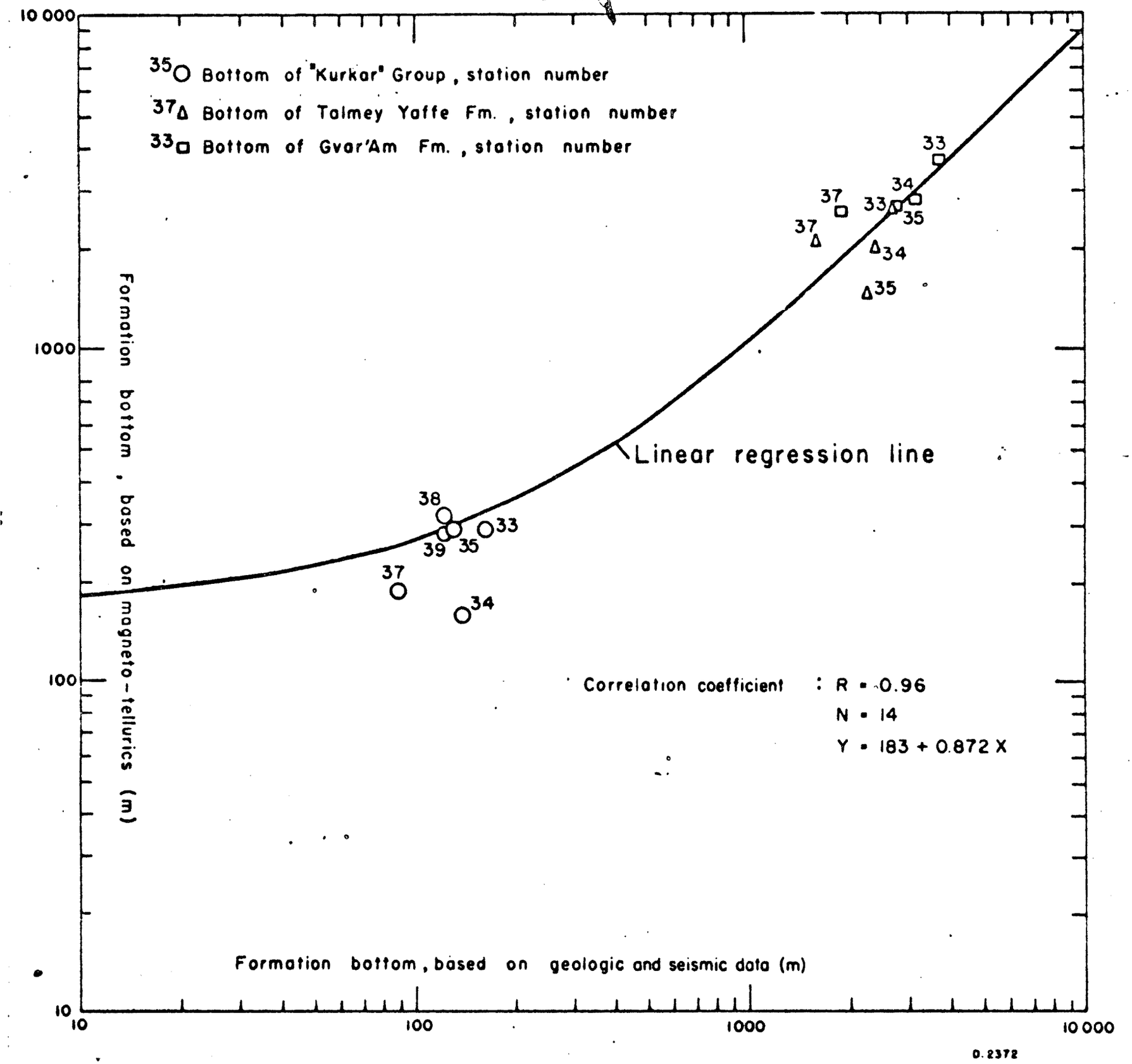

35 ○ Bottom of "Kurkar" Group, station number
37 Δ Bottom of Talmey Yaffe Fm., station number
33 □ Bottom of Gvar'Am Fm., station number
Linear regression line
Correlation coefficient : R = 0.96
N = 14
Y = 183 + 0.872 X
Formation bottom, based on magneto-tellurics (m)
Formation bottom, based on geologic and seismic data (m)
10 000
1000
100
10
10
100
1000
10 000
0.2372

Fig. 7

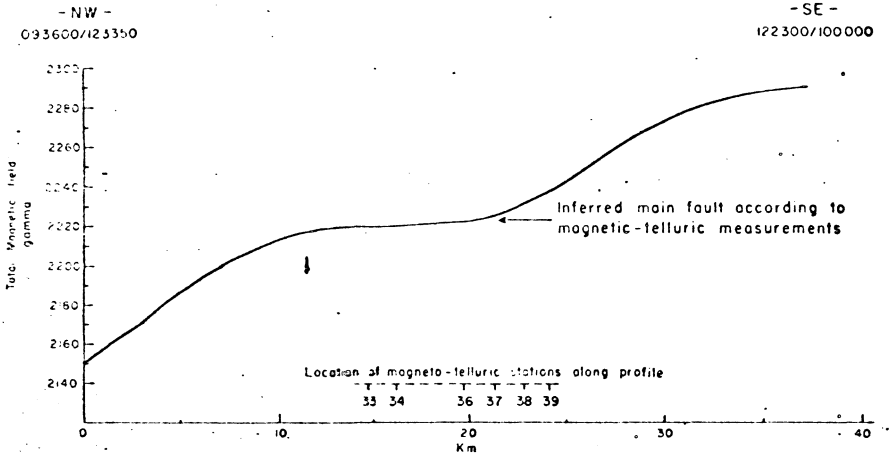
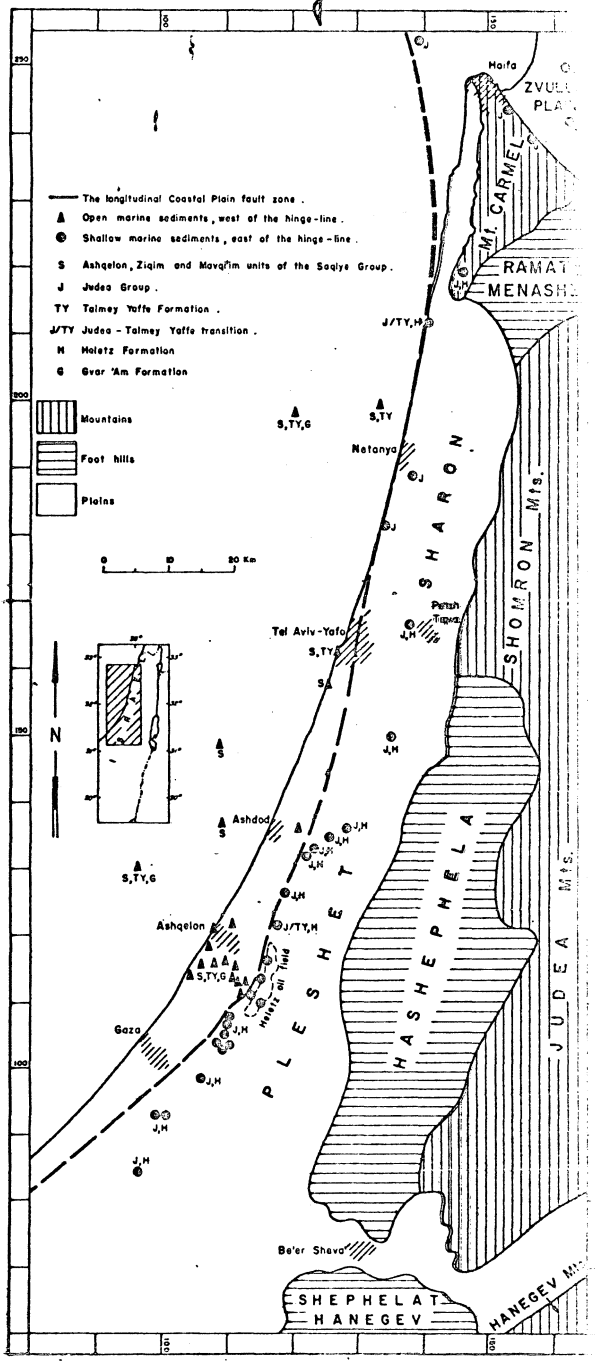

- NW -
093600/123350
- SE -
122300/100000
Total Magnetic Field gamma
2300
2280
2260
2240
2220
2200
2180
2160
2140
Inferred main fault according to magnetic-telluric measurements
Location of magneto-telluric stations along profile
33 34 36 37 38 39
0
10
20
30
40
Km

Fig. 8


The longitudinal Coastal Plain fault zone.
Open marine sediments, west of the hinge-line.
Shallow marine sediments, east of the hinge-line.
S Ashqelon, Ziqim and Mavqi'im units of the Saqiye Group.
J Judea Group.
TY Talmey Yafe Formation.
J/TY Judea - Talmey Yafe transition.
H Holetz Formation
G Gvar 'Am Formation
Mountains
Foot hills
Plains
0 10 20 Km
N
Haifa
ZVULUN PLAIN
MT. CARMEL
RAMAT MENASHE
SHARON
SHOMRON Mts.
Netanya
Petah Tiqwa
Tel Aviv-Yafo
Ashdod
Ashqelon
Gaza
Holeta oil field
PLESHET
HASHEPHELA
JUDEA Mts.
Be'er Sheva
SHEPHELAT HANEGEV
HANEGEV